INCH-POUND
MIL-DTL-32015/5(AS)
9 April 2004

DETAIL SPECIFICATION SHEET

MICROPHONES, LINEAR, CHEMICAL, BIOLOGICAL, RADIOLOGICAL (CBR), NON-OXYGEN MASK

This specification is approved for use by the, Naval Air Systems Command, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32015.

AMSC N/A FSC 5965

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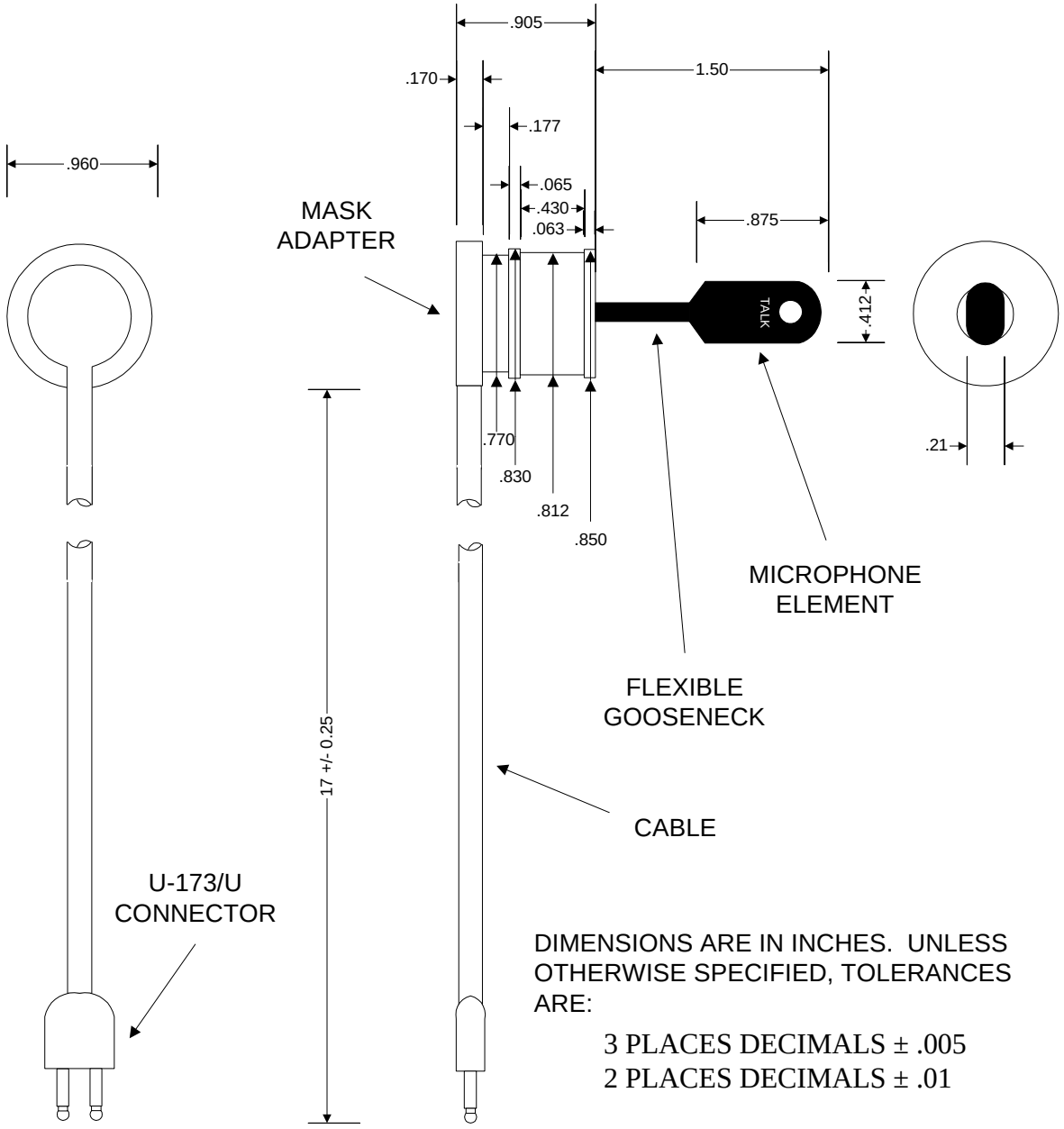


FIGURE 1. M32015/5-1 Microphone/Cable

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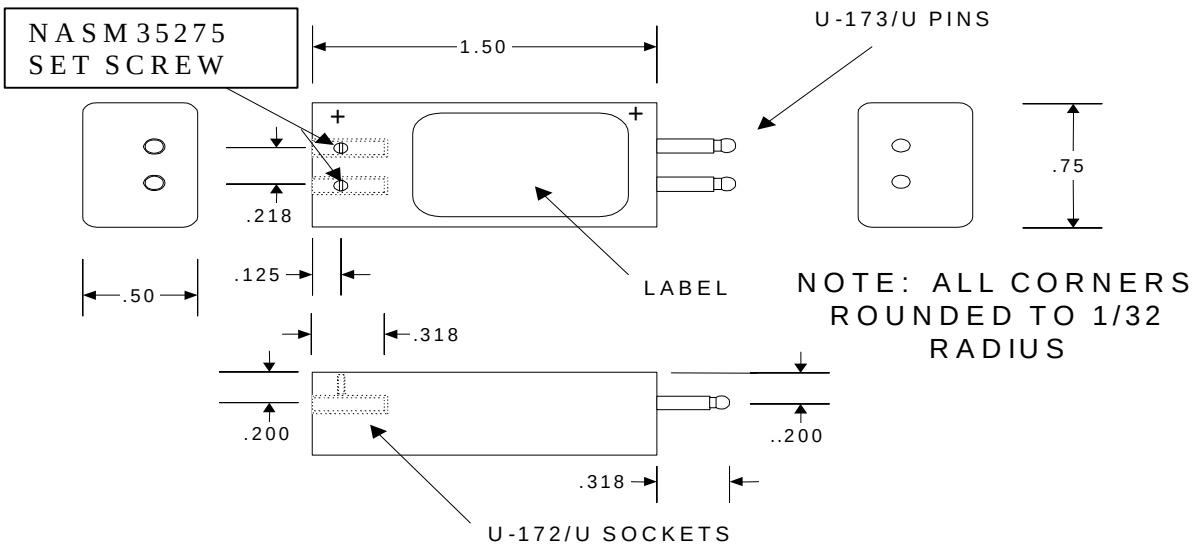


FIGURE 2. M32015/5-2 Microphone preamplifier.

APPLICABLE DOCUMENTS

NASM35275	Screw, Machine-Drilled Fillister Head, Slotted, Corrosion Resisting Steel, Passivated, UNC-2A

(Application for copies of NASC publications should be addressed to the Aerospace Industries Association, 1250 Eye Street NW, Washington DC 20005, or matthew@aia-aerospace.org.)

APPLICABLE DRAWINGS

AIR FORCE

57B12662	-	Connector, Plug, Electrical U-173/U
57B12664	-	Tip-Electrical Plug Connector

(Copies of the above drawings are available from Defense Supply Center Columbus, DSCC-VTCD, P.O. Box 3990, Columbus, OH 43216-5000, or www.charles,mcgrath@dla.mil.)

REQUIREMENTS:

DESIGN and CONSTRUCTION: See figures 1 and 2.

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CONFIGURATION: The M30215/5 microphone assembly is comprised of an integral microphone/cable subassembly (see figure 1) and a preamplifier subassembly (see figure 2). The microphone/cable assembly shall contain the microphone element, only the electronics necessary for transfer of the audio signal to the amplifier, and a flexible cable terminated in a U-173/U connector.

PIN RETENTION. As specified in the general specification.

COMPONENT PARTS: The M32015/5 microphone assembly shall be supplied as follows:

1 Microphone/cable unit, figure 1.
1 Preamplifier assembly, figure 2.

PART or IDENTIFYING NUMBERS (PIN):

M32015/5-1 Microphone/Cable Unit
M32015/5-2 Preamplifier Unit

WEIGHT: 62-g max. for the combined subassemblies.

COLOR: BLACK

CONNECTOR INTERFACE:

MICROPHONE/CABLE OUTPUT PINS: The output terminals shall be two pins made in accordance with Air Force Drawing 57B12664. The pins shall be arranged as detailed in Air Force Drawing 57B12662 and together shall comprise the U-173/U connector. The positive pin of the connector shall be marked with a “+” symbol on the connector.

PREAMPLIFIER INPUT SOCKETS: The preamplifier-input terminals shall accept the U-173/U connector of the microphone/cable. The preamplifier shall contain two 2-56 UNF slotted setscrews to secure the U-173/U plug of the microphone/cable subassembly. When tightened, the setscrews shall be recessed in the preamplifier body.

PREAMPLIFIER OUTPUT PINS: The output terminals shall contain two pins made in accordance with Air Force Drawing 57B12664. The pins shall be arranged as detailed in Air Force Drawing 57B12662 and together shall comprise the U-173/U connector.

AMPLIFIER BIAS POLARITY: The amplifier positive input and output pins shall be marked with “+” on the amplifier body.

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MASK INTERFACE: The microphone/cable subassembly shall be configured to install in the A/P22P-14 (V) x respirator assembly with the microphone element inside the mask. The microphone shall not poke the aircrew in the mouth/face or interfere with the aircrews drinking tube within the mask. Communications shall not be affected by small amounts of water droplets that may accumulate on the microphone element as a result of drinking water. The element shall be adjustable and shall be marked in contrasting color with "TALK" on the appropriate side.

INPUT CURRENT: 8-ma max. at 10 VDC when connected as shown on figure 3.
NOTE: All tests except overvoltage and reverse polarity shall use this test setup.

IMPEDANCE: The electrical impedance of the microphone assembly shall be such that the unit shall meet the sensitivity and frequency response to a close sound source while connected to a 150 ohm ±1 percent load impedance.

VOLTAGE OPERATING RANGE: 8 VDC to 12 VDC.

SENSITIVITY: 18mV(min) to 36mv(max) at 94 dB SPL with 10 VDC supply voltage using the normal test setup of figure 3.

SENSITIVITY AT REDUCED BIAS VOLTAGE: Shall not vary more than 3 dB at 104 dB SPL from the 10VDC established value when the supply voltage is varied between 8 and 12 volts.

SENSITIVITY AT ALTITUDE: ±3 dB of sensitivity at ground level while at a simulated altitude of 15,000 ft above sea level.

FREQUENCY RESPONSE AT GROUND LEVEL: The frequency response characteristic of the microphone, with moisture barrier in place, between 400 and 6,000 Hz, shall be within the response envelope shown on figure 4.

FREQUENCY RESPONSE AT ALTITUDE: Within ±5 dB of the frequency response at sea level while at a simulated altitude of 15,000 feet above sea level.

NOISE IMMUNITY: The noise-immunity characteristic of the microphone shall intersect the zero axis at 2,500 Hz minimum, and have a slope of at least 6 dB per octave (see figure 5).

LINEARITY: As specified in the general specification. The resistive load shall be 150 ±1 percent ohms.

HARMONIC DISTORTION: As specified in the general specification.

GAIN STABILIZATION: Within 100 ms when supply voltage is varied between 8 and 12 VDC.

OVERVOLTAGE PROTECTION: No damage to microphone or preamplifier when subjected to +28 VDC for 30 minutes using the overvoltage test circuit shown on figure 6.

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REVERSE POLARITY PROTECTION. No damage to microphone or preamplifier when subjected to -28 VDC for 30 minutes using figure 6 test circuit.

EXPLOSIVE ENVIRONMENT: N/A

RAIN: As specified in the general specification.

CHEMICAL AGENT PERMEATION: The materials of the microphone/cable external to the mask and the materials of the preamplifier shall be resistant to liquid agent permeation (mustard and thickened soman) as specified in general specification. The microphone/cable and preamplifier units' operation shall not be degraded by exposure to the chemical agents.

QUANTITATIVE FIT FACTOR (QFF). As specified in the general specification.

ENVIRONMENTAL STRESS SCREENING (ESS): 24 hours of temperature cycling between -20 and +85 °C, with each cycle consisting of 1 hour at each temperature and the change between temperature occurring at a minimum rate of 5 °C per minute. The microphone assemblies shall be functionally tested prior to each cold to hot transition and upon completion of the temperature cycling.

MASK PRESSURIZATION: N/A.

MARKING:

MICROPHONE/CABLE ASSEMBLY (Location of marking manufacturer's option)

M32015/5-1 **(10VDC)** (FOR V-22 AIRCRAFT ONLY) *
MANUFACTURER'S NAME OR CAGE CODE
CONTRACT NUMBER
DATE CODE
SERIAL NUMBER

PREAMPLIFIER ASSEMBLY

M32015/5-2 **(10VDC)** (FOR V-22 AIRCRAFT ONLY) *
MANUFACTURER'S NAME OR CAGE CODE
CONTRACT NUMBER
DATE CODE
SERIAL NUMBER

* Voltage designation shall be molded or engraved and ink-filled.

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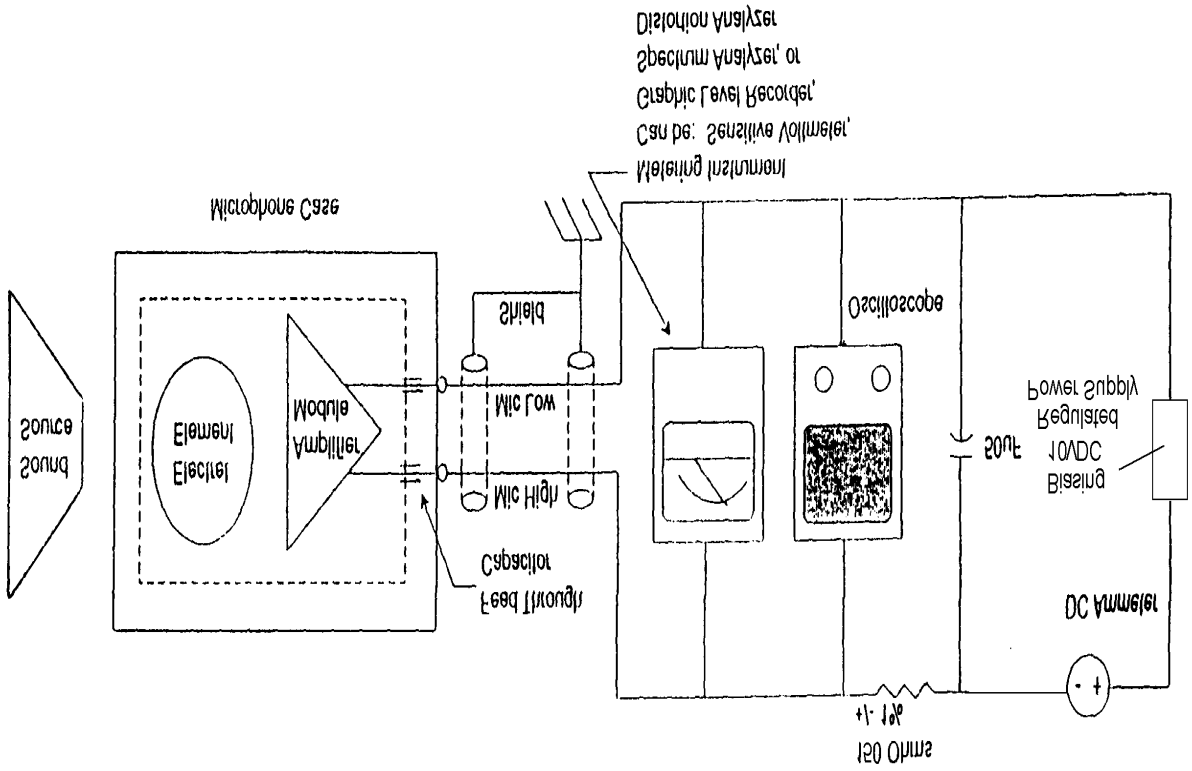


FIGURE 3. Test circuit schematic.

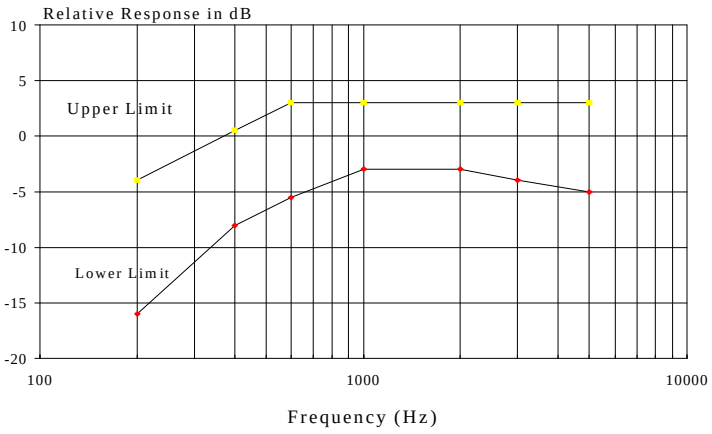


FIGURE 4. Frequency response limit curves

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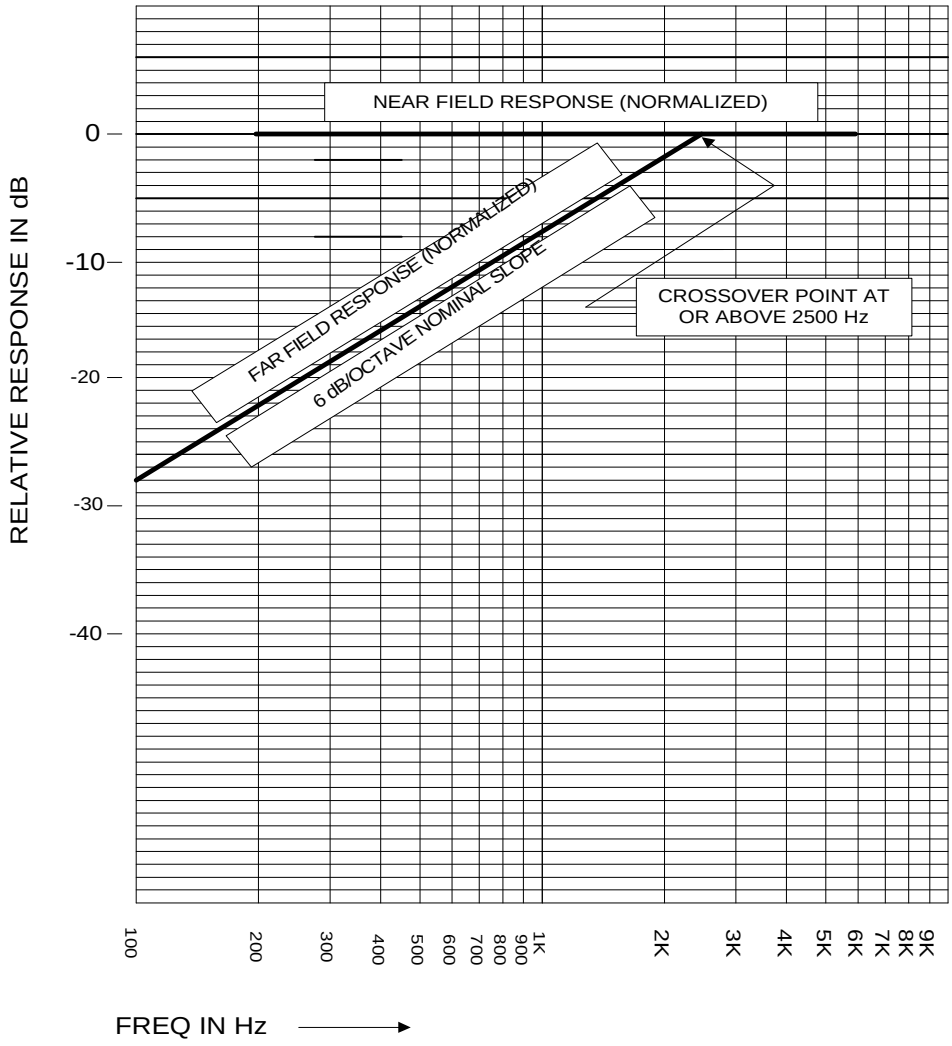


FIGURE 5. Noise immunity characteristics.

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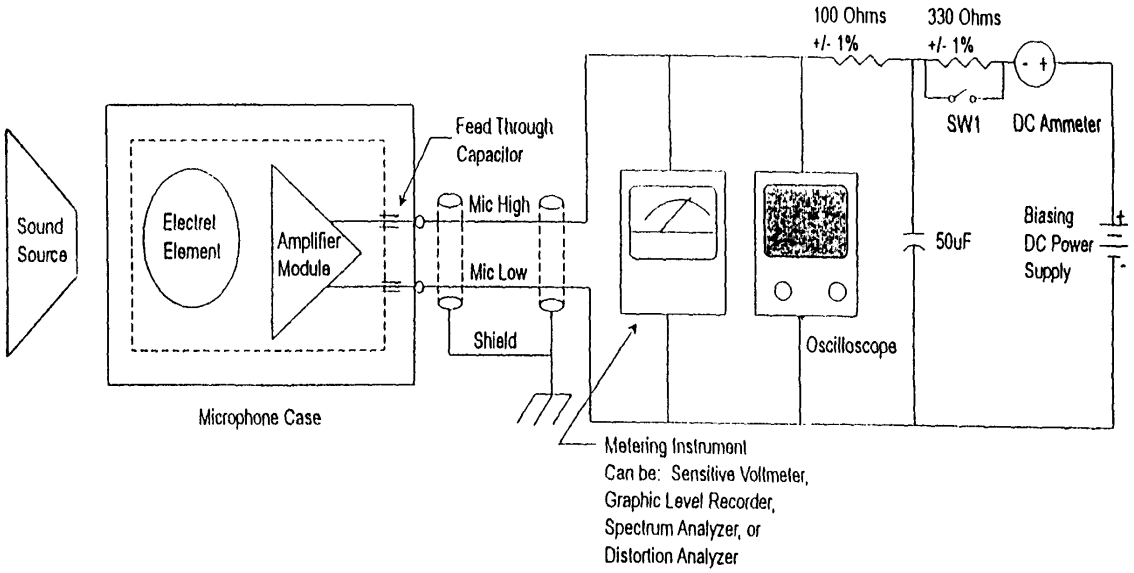


FIGURE 6. Overvoltage test circuit schematic.
(for overvoltage and reverse polarity tests only)

INTENDED USE: The microphone, M32015/5, is a noise-canceling, DC-powered, high-gain microphone designed for use on a non-oxygen mask in conjunction with the A/P22P-14 (V) x protective CBR assembly used by V-22 Osprey tilt-rotor aircrewmen, on the ground or in the air, or against CBR warfare agents by aircrewmen equipped with a battery-powered, body-worn portable intercom unit. The microphone is intended for use as a part of the airborne communication system that will provide communication of high intelligibility under the extreme noise conditions encountered in military aircraft. The microphone is designed to work with 10-VDC biased intercommunication systems.

Preparing activity:
Navy - AS
(Project 5965-0397)

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